

OTTUMWA, IA, USA

WMO: 725465

Lat: 41.108N

Lon: 92.447W

Elev: 842

StdP: 14.25

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14950

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.4	(c) 0.6	(d) -11.9	(e) 3.0	(f) -3.3	(g) -7.7	(h) 3.8	(i) 1.6	(j) 31.2	(k) 20.5	(l) 28.2	(m) 18.7	(n) 10.0	(o) 300	(p) 0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	92.0	76.4	89.5	75.4	86.9	73.8	79.0	88.4	77.3	86.2	75.6	83.9	11.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.2	141.2	85.0	74.6	133.4	83.2	72.9	126.0	81.2	43.1	88.5	41.3	86.2	39.7	84.1	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.6	(b) 23.9	(c) 20.7	(e) -11.9	(f) 96.5	(g) 5.9	(h) 4.0	(i) -16.1	(j) 99.4	(k) -19.6	(l) 101.7	(m) -22.9	(n) 104.0	(o) -27.2	(p) 106.9		
(4) 26.6	(5) 23.9	(6) 20.7	(7) -11.9	(8) 96.5	(9) 5.9	(10) 4.0	(11) -16.1	(12) 99.4	(13) -19.6	(14) 101.7	(15) -22.9	(16) 104.0	(17) -27.2	(18) 106.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	23.1	27.4	39.4	51.4	62.0	71.6	75.3	73.1	65.8	53.3	40.1	28.4
	DBStd	20.54	12.54	12.60	11.74	9.77	8.36	6.17	5.82	5.46	8.12	9.66	10.51	11.60
	HDD50	3019	833	637	367	97	8	0	0	0	2	76	326	673
	HDD65	6095	1298	1053	794	418	158	16	2	6	88	382	746	1134
	CDD50	3395	0	3	39	140	380	647	783	717	476	177	30	3
	CDD65	1001	0	0	2	10	66	213	321	258	113	18	0	0
	CDH74	9013	0	1	13	132	569	1845	3116	2179	1004	151	3	0
	CDH80	3133	0	0	1	25	136	620	1240	754	327	30	0	0
Wind	WSAvg	10.5	11.9	11.9	12.3	12.6	10.9	9.3	8.1	7.6	8.7	10.3	11.3	11.1
Precipitation	PrecAvg	35.5	1.1	1.1	2.4	3.6	4.3	4.4	4.4	4.2	4.0	2.6	2.0	1.3
	PrecMax	54.5	3.3	3.8	5.1	7.0	9.4	11.2	11.2	9.8	13.7	6.6	5.5	3.7
	PrecMin	22.5	0.1	0.2	0.4	1.1	0.7	0.8	0.8	0.4	0.5	0.0	0.1	0.1
	PrecStd	7.4	0.7	0.7	1.5	1.6	2.1	2.4	2.4	2.4	2.6	1.6	1.4	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	63.9	75.5	83.8	88.8	92.2	97.5	94.8	91.2	84.0	72.7	61.7
		MCWB	47.7	53.1	61.8	64.6	72.9	76.3	78.3	74.8	74.0	68.9	58.6	56.4
	2%	DB	50.1	56.8	69.4	77.8	83.6	89.6	93.1	90.4	87.6	78.4	66.0	54.6
		MCWB	43.2	48.3	57.9	62.0	69.0	74.9	77.3	75.9	72.7	64.6	55.8	48.8
	5%	DB	44.5	50.2	63.2	73.0	80.2	86.8	90.1	87.3	83.9	73.3	60.8	48.8
		MCWB	39.8	43.8	54.3	59.1	66.4	73.2	76.6	74.6	70.4	62.0	52.6	43.8
	10%	DB	39.0	44.3	57.4	68.3	76.7	83.6	86.8	84.1	80.0	69.2	56.1	43.6
		MCWB	35.3	39.5	49.2	56.5	64.8	71.9	74.7	73.1	68.1	60.6	49.2	39.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.9	55.8	63.9	67.8	74.5	79.4	81.5	80.4	76.8	71.7	61.7	57.8
		MCDB	54.2	61.7	72.9	79.2	84.7	88.6	91.4	88.6	87.5	79.9	69.5	60.7
	2%	WB	44.0	49.1	59.8	64.0	71.7	77.1	79.5	78.3	74.7	67.9	57.7	50.4
		MCDB	48.9	56.1	66.0	74.3	81.3	86.2	89.7	86.7	83.9	74.1	64.2	53.3
	5%	WB	39.6	44.3	55.3	61.1	69.2	75.1	77.9	76.4	72.7	64.6	53.9	44.0
		MCDB	44.6	50.2	62.7	70.8	77.1	83.8	87.2	84.5	80.6	71.2	59.8	48.1
	10%	WB	35.7	39.3	49.8	58.0	66.8	73.3	76.2	74.6	70.4	61.4	49.8	39.5
		MCDB	38.8	44.1	56.9	66.5	74.1	81.4	84.6	82.0	77.5	68.4	55.9	43.2
Mean Daily Temperature Range	5% DB	MDBR	16.1	16.6	19.1	21.1	19.5	18.7	19.2	19.4	21.7	20.3	18.2	15.3
		MCDBR	22.8	24.3	26.4	27.5	22.7	21.2	21.2	22.4	24.0	24.5	24.5	21.3
		MCWBR	17.1	16.9	16.2	15.6	11.7	10.4	9.4	10.6	11.0	13.5	16.1	16.3
	5% WB	MCDBR	21.8	22.7	23.8	24.0	20.1	18.8	19.2	19.3	20.7	20.3	22.2	19.9
		MCWBR	17.0	16.6	15.6	15.3	11.6	10.9	9.9	10.8	10.7	13.5	16.3	16.4
Clear-Sky Solar Irradiance	taub	0.293	0.309	0.335	0.387	0.413	0.420	0.438	0.420	0.387	0.342	0.299	0.290	
	taud	2.417	2.371	2.390	2.261	2.255	2.281	2.219	2.287	2.331	2.458	2.507	2.446	
	Ebn at Noon	274	286	290	280	274	272	265	267	269	271	271	266	
	Edh at Noon	25	31	34	42	43	42	44	40	36	28	23	23	
All-Sky Solar Radiation	RadAvg	603	915	1195	1508	1766	1959	1968	1723	1474	984	666	508	
	RadStd	61	71	102	112	127	198	141	100	107	111	57	51	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page